

Scientific rationale

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Hashes bind exact inventory/control bytes, not a PDF containing its own digest. The source-inventory scopes and reproducible build recipe are delivered with this document. Document verification supplies no numerical conformance, qualification, validation, readiness or production authority.

1 What this conditional core makes precise

FRUIT separates the measured input, a feedback model, a residual-processing path and a model-rejoin path. A model can alter what enters residual processing and contribute to the final product. The scientific meaning of that product therefore depends on the exact path composition, its information sources and what was fixed or learned along the way.

This revision makes those dependencies explicit without choosing a numerical procedure. Program-order authority permits authorship; conditional-composition authority permits deductions under stated premises. Numerical-method authority is a separate decision. A conditional identity can be exact while every numerical realization remains unavailable.

The sole current review-candidate normative core is SCI-FRUIT-NORMATIVE-CORE v0.1/r0.4, bound by the identical source-inventory digest on this cover and the conformance cover. This rationale explains selected consequences. It does not add requirements, methods or evidence rules. The complete assumptions, requirements, predictions and product/state definitions remain in the core.

The frozen Stage A packet and eleven admitted payloads remain unchanged. The r0.3 fixed-operator repair and three-document structure are accepted. The r0.4 directive restores only the admitted analysis/gridding-coefficient ownership and route-use boundary. Internal modal application coordinates remain separate from those downstream handoff facts. The accepted response specialization and other scientific rules are unchanged. The r0.4 core is the current review candidate, without hidden inherited obligations, owner approval, freeze, Registry registration or activation. Role names are core declarations only.

A later numerical method must resolve the actual parent route, model, projection, residual processing, rejoin, support, learning, recurrence, stopping and selector. Upstream producer permissions remain necessary. Neither this document nor its engineering companion fills those choices.

Normative source: review-candidate core, Authority and typed objects; REQ-001, REQ-023–024.

2 Three model roles and the applied operand

A candidate model is proposed. An accepted model is selected for feedback. The applied model is the exact object used by removal and rejoin. Their spaces may differ, and their relationship is a scientific transformation rather than an implied equality.

Normalization, clipping, sign conversion, support restriction, projection or unit conversion can change the applied object. An accepted-model uncertainty or error therefore cannot be placed directly on applied-model axes without that transformation and its dependence. Selection and transformation variation also require separately addressable uncertainty roles.

The conditional composition is imported directly from the normative source:

$$\rho_k = y_k^{\text{in}} - \Pi_k(m_k^{\text{applied}}), \quad z_k = F_k(\rho_k) + B_k(m_k^{\text{applied}}). \quad (1)$$

Here $\Pi_k(m_k^{\text{applied}})$ is projected removal, $F_k(\rho_k)$ is the complete processed residual and $B_k(m_k^{\text{applied}})$ is rejoin. The method defines every operation location, bypass and application count. There is no assumed inverse, identical projection/rejoin, equal input/output space or universal bypass.

Removal and rejoin each require exact compatible quantity, unit/scale, frame, domain, sampling, support, additive origin/reference or gauge, null-space, calibration and response conventions. Affine or quotient representations need well-defined arithmetic in the declared reference; selecting arbitrary representatives can change the result. Equal array dimensions cannot establish this compatibility.

A feedback model is not thereby truth, a source detection, posterior or a Pointing/OOF result. A rejoined unavailable measured mode remains model/prior supported. The scientific question is which path supplied that content, not whether a finite number appears in the final array.

Normative source: review-candidate core, Typed spaces and Composition; REQ-002–005, REQ-015.

3 A hierarchy of exact equations

The full composition is the starting point. Only a fixed linear complete residual procedure permits distributive expansion:

$$z_k = A_k y_k^{\text{in}} - A_k [\Pi_k(m_k^{\text{applied}})] + B_k(m_k^{\text{applied}}). \quad (2)$$

This remains map-valued if projection or rejoin is nonlinear. Only complete fixed linearity of all three maps permits the linear applied-model operator:

$$D_k := B_k - A_k \circ \Pi_k, \quad z_k = A_k y_k^{\text{in}} + D_k m_k^{\text{applied}}. \quad (3)$$

A fixed structural basis alone is insufficient. Centering, operator-defining resolution rules, support and all consequential state belong to the premise. Affine offsets remain unless complete linearity is actually established.

The hierarchy prevents a convenient linear expression from silently taking over a nonlinear or adaptive procedure. The limiting cases are conditional: zero applied feedback reduces to ordinary residual processing only for zero-preserving removal/rejoin; identity residual processing cancels removal and rejoin only when those contributions and spaces agree. If $D_k = 0$ under its complete premises, the fixed result is independent of the applied model. None of these deductions chooses a numerical operator or proves recovered sky.

For an applied-model difference under the same fixed operators/input, the result difference is $D_k \delta m_k^{\text{applied}}$. The operator is model-path transfer; the result difference is the distinct model-induced output shift. Model mismatch and bias have their own identities; a deterministic difference is not covariance. Calling its expected shift bias needs one exact estimand, reference/truth model, error random variable, probability law and conditioning. Without those premises, bias is unavailable even when the shift is well defined. Accepted error must first be transformed into applied space.

Normative source: review-candidate core, Composition and model-error transfer; ASM-001-004, REQ-006 and REQ-015.

4 What a response query holds fixed

RF-01 perturbs iteration input; RF-02 perturbs the applied feedback model. Both freeze complete operator-defining state and the complete induced numerical maps, including support, reference/gauge/null-space, branch and the other operand. Internal coordinates that evaluate those maps may change with the perturbed operand. A changed modal coordinate is not by itself a changed map.

For differentiable fixed maps on compatible tangent spaces:

$$\begin{aligned} J_{z \leftarrow y, k}^{\text{fixed}} &= DF_k|_{\rho_k}, \\ J_{z \leftarrow m, k}^{\text{fixed}} &= DB_k|_{m_k^{\text{applied}}} - (DF_k|_{\rho_k}) \circ (D\Pi_k|_{m_k^{\text{applied}}}). \end{aligned} \tag{4}$$

Bind baseline, units/basis, support, codomain, derivative convention and fixed conditioning. Complete fixed linearity reduces these to A_k, D_k ; a fixed affine operation still retains its additive output term. A failed derivative or domain/comparison failure remains unavailable or a separately typed finite-difference/transition query, never silently RF-03.

RF-03 reruns prescribed operations that can change defining maps or consequential state: construction/acceptance, loading subspace, metric, support/mask, centering/reference, rank, grouping, threshold, selection or branch. Prior iteration state and upstream procedure/state remain fixed. The distinction is causal: reevaluating an internal coordinate under a fixed map does not rerun its defining-state resolution. An unresolved particular map remains unavailable; calling an internal coordinate an application coefficient does not change its role.

RF-04 propagates a fixed iteration extent. RF-05 includes stopping and terminal selection. RF-06 reruns the included upstream chain. RF-07 fixes upstream producer state/generation and composes the complete FRUIT-only parent-domain query through the applicable local/recursive/terminal components. Equal values cannot make these query scopes interchangeable.

Normative source: review-candidate core, Operator roles; both response sections; ASM-005; REQ-011–013.

5 Why fixed-subspace PTC remains one map

PTC owns its estimator and exact fixed-state application map; FRUIT owns only the response role assigned to a declared use of that operation. The exact admitted group/time equations yield the following shared core relation:

$$\begin{aligned} G_{g,t}^{\text{app}} &= \hat{A}_g^T W_{g,t}^{\text{app}} \hat{A}_g, & \hat{m}_{g,t}^{\text{app}}(x) &= x_{g,t} W_{g,t}^{\text{app}} \hat{A}_g (G_{g,t}^{\text{app}})^+, \\ P_{g,t,\Theta}^{\text{PTC}} &= I - W_{g,t}^{\text{app}} \hat{A}_g (G_{g,t}^{\text{app}})^+ \hat{A}_g^T, & & \\ z_{g,t} &= x_{g,t} - \hat{m}_{g,t}^{\text{app}}(x) \hat{A}_g^T = x_{g,t} P_{g,t,\Theta}^{\text{PTC}}, & \delta z_{g,t} &= \delta x_{g,t} P_{g,t,\Theta}^{\text{PTC}}. \end{aligned} \quad (5)$$

Here every symbol is PTC-local. Frozen loading, metric, normal matrix/inverse, centering, support, detector binding, rank/tolerance, mask, orientation, branch and other defining facts fix P . The modal coordinate $\hat{m}^{\text{app}}(x)$ changes algebraically with x while P stays fixed. This is fixed-map evaluation, not relearning or a new operator generation. The full FRUIT response still composes its actual remove/process/rejoin paths.

Fixed centering of uncentered calibrated input gives:

$$\begin{aligned} x_{g,t} &= Y_{g,t}^{\text{CAL}} - \lambda_g, \\ z_{g,t} &= (Y_{g,t}^{\text{CAL}} - \lambda_g) P_{g,t,\Theta}^{\text{PTC}}, \quad \text{D}_Y z[H] = H P_{g,t,\Theta}^{\text{PTC}}. \end{aligned} \quad (6)$$

The output is affine; its fixed term $-\lambda_g P$ is retained. Its derivative is HP . A separately named frozen numerical-component subtraction instead holds $\hat{U}$ fixed in $Y' - \hat{U}$ and has identity derivative H . Those are distinct operations even if one result happens to agree.

The perturbation must remain in the source's exact finite/rank/support domain. Failure produces affected unavailability or a separately typed transition. Rerunning a defining-state rule is a procedure response. Neither outcome changes the PTC estimator, chooses rank/support or authorizes a numerical FRUIT method, route or fidelity claim.

Normative source: review-candidate core, Exact admitted PTC specialization; PRED-006; REQ-012–013.

6 Support and information sources remain visible

Support is not one universal mask. Measured-parent support, the three model stages' definition support, removal support, residual-processing support, rejoin support and final-output support describe different scientific roles. Learning influence records transitive dependence; response-query support records the exact perturbation and result domain. All eight remain separately addressable even when some happen to coincide.

The method determines whether compatible residual-and-model, residual-only or model-only composition is permitted. Missing information is not a numerical zero. Undefined unequal-support behavior is unavailable, with affected scope and cause. No implicit fill, extrapolation, renormalization, reference offset or gauge alignment is supplied by the composition notation.

Occurrence class and mode origin answer different questions. A location can have both paths available while a particular absolute/additive mode is supported only by model or prior. Rejoin then adds model-sourced content; it does not turn that mode into an observationally measured quantity. A reference choice similarly supplies coordinates, not an observation with zero uncertainty.

Iteration does not create exposure or independent measurements. Reusing a parent, shared learned model, calibration, response or reference retains dependence. Iteration count therefore supplies no square-root uncertainty scaling. Observation-local, coadd-level and joint learning also cannot be interchanged merely because output shapes match.

These distinctions preserve upstream ownership. In particular, the inherited JINC base product does not gain a coadd, covariance, significance or response product because the FRUIT core names a conditional obligation for one. Unavailable producer authority stays unavailable for its dependent claim.

Normative source: review-candidate core, Ownership; Path support and information origin; REQ-007–009, REQ-022.

7 Law-relative uncertainty and NOI member meaning

A stochastic target names its exact law, random variables, fixed objects, resolution/selection condition, omitted variation and scientific domain/reference. Pre-outcome means before that law's random event. Physical-parent uncertainty keeps parent/model values random. Deliberately fixing the realized model defines a narrower target with learning omitted. A conditional NOI-assignment law can instead fix realized parent/model/operator/support and exact successful design resolution before drawing its member. These targets are not equivalent.

Under complete fixed linearity and finite compatible joint moments, the law-qualified parent/applied covariance has all four terms:

$$\begin{aligned} C_z^{\text{FRUIT}} = & A_k C_{yy}^{\text{FRUIT}} A_k^\top + D_k C_{mm}^{\text{FRUIT}} D_k^\top \\ & + A_k C_{ym}^{\text{FRUIT}} D_k^\top + D_k C_{my}^{\text{FRUIT}} A_k^\top. \end{aligned} \quad (7)$$

Both cross terms remain unless exact law/conditioning justifies omission. They vanish with model variance only inside the explicitly fixed-model target. FRUIT branch/completion/convergence/terminal-conditioned targets need separate method and estimand authority; a generic conditioning symbol supplies none.

With fixed applied model/state, role- and iteration-qualified members obey

$$\begin{aligned} z_{k,b}^{\text{full}} &= A_k y_{k,b}^{\text{full}} + D_k m_k^{\text{applied}}, \\ \mathbb{E}_{\mathcal{L}}[z_{k,b}^{\text{full}} \mid \Omega_{k,\text{pre}(\mathcal{L})}^{\text{FRUIT}}] &= A_k \mathbb{E}_{\mathcal{L}}[y_{k,b}^{\text{full}} \mid \Omega_{k,\text{pre}(\mathcal{L})}^{\text{FRUIT}}] + D_k m_k^{\text{applied}}, \\ \mathbb{E}_{\mathcal{D}_{k,g}^{\text{NOI}}}[\delta y_{k,b}^{\text{NOI}} \mid \mathcal{H}_{k,g}^{\text{NOI}}] &= 0, \quad \delta z_{k,b}^{\text{NOI}} = A_k \delta y_{k,b}^{\text{NOI}}. \end{aligned} \quad (8)$$

The NOI zero is an expectation under its exact design law, not a demand that an individual member have zero spatial/sample mean. No finite-ensemble mean subtraction or $B - 1$ correction follows; source/residual structure may remain. Do not add deterministic rejoin to each zero-target-law member. A full-affine method needs its own exact center and estimator authority.

The interim owner directive supplies this conditional distinction. Actual external NOI law/product/estimator binding remains unavailable until supplied; the notation does not admit a numerical route. Candidate, selection, accepted, transformation, applied and cross-covariance roles remain independently addressable, alongside core-declared external/operator/support/selection/empirical roles. Bias and output shift are distinct products outside that family. Fixed-state, dependent-successor and relearned NOI remain different methods and ensembles, with omissions and immutable ancestry explicit.

Normative source: review-candidate core, Law-relative covariance; NOI and core-declared uncertainty roles; REQ-014–016.

8 Iteration, continuation and terminal selection

The core identifies transition roles without choosing recurrence. The method must specify whether an iteration consumes the original parent, a previous result, a reconstruction or another exact product. Every route retains immutable original measured-parent ancestry. Candidate, accepted, applied, cumulative model, diagnostic difference and result retain their own meanings.

Unchanged defining-state rules can generate new operator-defining coefficients, learned states and model values. Those changes require new realized generations, not a new method each time. Internal fixed-map coordinate evaluation alone creates no new operator generation. Changing the learning, support, recurrence, stopping or selection rule changes method identity. Ordered operations and exact bypass/application counts are part of that distinction.

After completed absolute zero-based iteration N , exact continuation starts at $N + 1$ from S_{N+1} . It requires all future-consequential scientific, stochastic and control state under the same compatible future controls and numerical profile. A saved result array does not prove sufficiency. Restart creates a new lineage; retry retains exact attempt and state accounting.

Completion, resource exhaustion, convergence, stopping and terminal selection are separate claims. A resource limit does not prove convergence, and a converged sequence does not establish scientific adequacy. No last, latest, lowest-residual or visually favorable iteration is selected by default.

A selector compares completed candidates in common quantity, units, frame, domain, gauge, response, support and uncertainty/metric meaning. If conversions are needed, they and the comparison belong to its method identity. The record retains every candidate, conversion, inclusion/exclusion, tie and unavailable comparison. Selection creates an immutable relation to a completed bundle; it never rewrites the selected iteration.

Normative source: review-candidate core, Operation state; Completion and terminal comparison; REQ-010–011, REQ-017–020.

9 Local completion and the present claim ceiling

Each request generation is immutably not-requested or requested; a later request creates a new generation. For a requested generation, the successful producer order distinguishes effective action, resolved scope, invocation, realized path values/outcomes, complete candidate, the FRUIT-owned decision, completed iteration and publication. Availability and producer outcome are separate branch/role records at the stage reached; early stop fabricates no later success. The result is a child payload: scientific consumption requires its exact bundle or immutable bundle reference.

A complete bundle supplies all universal and method-required roles, or the precise typed statuses allowed for narrower claims. A method cannot waive core roles or hard dependencies. Numerical result, continuation, response, uncertainty and terminal selection each retain the inputs and premises needed for their own claim. An unavailable optional companion does not erase an otherwise permitted narrower local result.

Terminal-use references are open-world: record only requested/linked decisions or an explicit no-request statement. For SCI-VAL use retain exact request, applicability, eligibility and realization with profile/source binding. Not-requested creates no eligibility proposition. Missing profile/scope can produce a realized accounting record with applicability_unknown and decision_unavailable; incomplete/failed/not-produced artifacts are separate realization conditions. These child decisions do not erase base completeness or create a FRUIT iteration. Only an explicitly requested narrower product role may require a child. SCI-VAL evaluates use policy, not FRUIT execution, model choice or terminal selection.

The engineering specification explains how a future exact candidate would produce requirement-level evidence. It supplies no numerical conformance now. No numerical method, route, response/covariance fidelity, recovered measured mode, coverage, convergence adequacy, observational validation, performance, readiness, downstream fitness or production authorization is claimed.

The next decision is Grant Wilson's separate review of this revised conditional core and possible conditional freeze. Numerical development, qualification and policy recommendations remain separately authorized work with inherited independent-pointing evidence obligations. The present documents complete conditional authorship and document verification only.

Normative source: review-candidate core, Products and lifecycle; Universal completion and assumptions; REQ-018, REQ-021-024.